

DUBLIN BUSINESS SCHOOL



ADVANCED DATABASES

B9IS125

Advanced Databases Continuous Assessment: Database Implementation Report

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Abstract

This report outlines the structure of the database created from the database designed in the first part of the project, designed to serve a public library, as well as SQL triggers, views and stored procedures created for the well functioning of the database, realistic test data created to populate the database, and possible future innovations that would improve the current functioning of the database.

Contents

1	Introduction	2
2	Implementation	3
2.1	Database Creation	3
2.2	Database Population	5
2.3	Triggers	6
2.4	Views	7
2.5	Procedures	7
3	Conclusion	8
3.1	Inovation	9

1 Introduction

As a continuation to the development of the database, which was designed in phase 1 of the project, the database was now implemented in a Microsoft SQL Server.

The database, as outlined in the previous report, was designed for a public library containing a large collection of books, employees, each furnished with an username and password required to access the system, and serving members of the public, who do not require a subscription, but must register with the library so to be reachable in case there are problems with the devolution of a rented book.

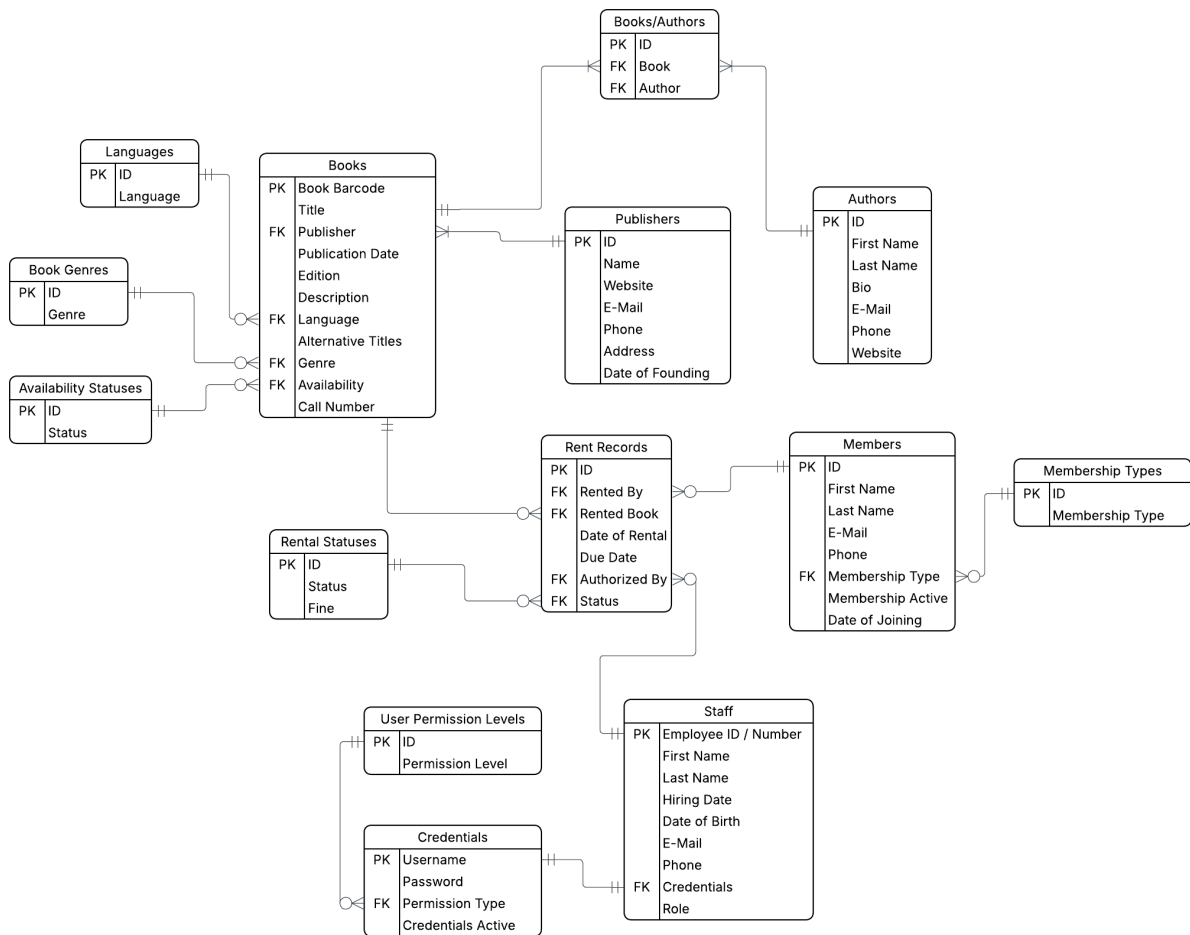


Figure 1: ERD diagram of the database, demonstrated on phase 1.

2 Implementation

2.1 Database Creation

The database was created under the name **Library**, and tables were then created following the original design, as seen in fig. 1. However, due to the unforeseen fact that some of the names used for columns in the design happen to be SQL reserved words, the names of certain fields had to be changed in order to avoid conflict with these keywords.

- **Languages:** The field "Language" had to be renamed to "BookLanguage".
- **BookGenres:** The table "BookGenres" could be created according to specification, with no changes.
- **AvailabilityStatuses:** As "Status" is a reserved word, the field was renamed to "AvailStatus".
- **RentalStatuses:** As with the "AvailabilityStatuses" table, the column "Status" had to be renamed to "RentStatus".
- **MembershipType:** The table "MembershipType" was able to be created according to specifications.
- **UserPermissionLevels:** The table "UserPermissionLevels" was, fortunately, also able to be created according to previous specifications.
- **Publishers:** The columns "Name" and "Address" both had to be renamed, to "PublisherName" and "PublisherAddress" respectively. Moreover, during the database design phase, it was assumed that contact information for publishers would be readily available, however, that turned out not to be the case. Many publishers turned out to have no public-facing phone number or e-mail. Perhaps this would not be a problem in a real life situation, where the library would have means to contact associated publishers, however, within the constraints of this project, that was not the case. Therefore, the "not null" constraint (and, consequentially, the "unique" constraint as well) had to be removed from the "Publishers" table, in order to allow those whose phone numbers and/or e-mail addresses were not easily obtainable. The phone number field for "Publishers" and future tables also had to be expanded to comport 25 characters, in order to allow international numbers.
- **Author:** While no fields had to be renamed, similarly to the "Publishers" table, it turns out that contact phone numbers and e-mails were not as available as initially

imagined, and thus the "not null" and "unique" constraints had to be removed in order to allow for those whose contact details were not accessible.

- **Members:** Besides the phone number field expansion, the only other change necessary on the "Members" table was changing the column type of BOOL to the SQL Server-compatible BIT type, containing an option of 0 or 1 instead of TRUE or FALSE.
- **Credentials:** On the "Credentials" table, the field "Password" had to be renamed to "UserPassword" and the "CredentialsActive" field had to be changed from BOOL to BIT, similarly to the case described above, but otherwise could be created according to definitions.
- **Staff:** The "Staff" table could mostly be created to specifications, except for the field "Role", which had to be renamed to "EmployeeRole".
- **Books:** On the "Books" table, the field "Edition" had to be renamed to "BookEdition", the field "Description" to "BookDescription" and the field "Availability" to "AvailabilityStatus". Also, the field "CallNumber" had to be extended to 20 characters, considering the length of some of the Dewey Decimal System codes found in the population of the database.
- **RentRecords:** The column "Status" had to be renamed to "RentStatus", and the type for the column "DueDate" had to be changed from DATETIME to DATE in order to perform DATE operations.
- **BooksAuthors:** Named "Books_Authors" in the database, but otherwise, unchanged from specifications.

Other than individual changes on tables, an overarching change was the definition of INTEGER fields. SQL Server does not support INTEGER(num) type declarations, with num being the maximum number, therefore, all integer fields had to be adapted to one of the SQL Server native types, that is TINYINT, SMALLINT, INT and BIGINT. Considering that, tables with a limited/pre-determined number of entries, such as AvailabilityStatus and Language, were assigned TINYINT Primary Key IDs, as no table of this sort was expected beyond 255 entries. Other, continuously growing tables, such as Members and RentRecords, were given the default sized INTEGER type.

2.2 Database Population

For some tables, the entries were already defined during database design. The membership type in `MembershipType` must be only "Child", "Young Adult", "Adult", "Student", "Professor", "Researcher" or "Visitor". The availability status in `AvailabilityStatuses` must be only "Available", "Rented", "On Hold" or "Unavailable for Renting". The rental status in `RentalStatuses` must be only "On Hold", "On Time", "Overdue", "Returned" or "Book Lost". Public libraries in Ireland have been no longer charging fines on delayed books since January 2019 (*Do I Have to Pay Fines If I Return Items Late?* 2022), thus a generic 50 cents a day for late books and 10 Euro single fine were created.

User permission levels had not been defined beforehand, but should coincide with staff roles, thus were defined based on the operations expected for each staff member to make, such as only search for an assistant librarian, the ability to rent and return, but not register a new book, for a normal librarian, complete control of the collection for the curator, etc.

The languages with largest numbers of speakers were chosen to populate the `Languages` table (*List of Languages by Total Number of Speakers* 2026), as the more speakers there are, the more books are likely to be written in that language.

The books and genres used to populate the database were decided concurrently. A few genres were decided first, and it was decided that 5 books of each genre would be used to populate the database, but new genres were raised or dropped during the process. Books for each genre were located using the GoodReads (*Goodreads* 2026) website, which lists a massive number of books, and so were informations on the book description (or "blurb") and awards won. For each book, a generic Code-39 barcode was generated (*Code 39 Barcode Generator* 2026), and the website LibraryThing (*LibraryThing / Catalog Your Books Online / LibraryThing* 2026) was used to look up the Dewey Decimal System classification of each book to populate the call number.

With the books decided, authors and publishers were populated based on the books chosen, using publicly available information to populate contacts, websites, locations, etc. For authors, the website LibraryThing was also used to collect biographies ("bios") on each author, as well as a list of awards.

For members and staff members, names were created using publicly available name databases (*The Meaning and History of Surnames - Behind the Name* 2026 and *The Meaning and History of First Names - Behind the Name* 2026), as well as an online phone number generator (*Phone Number Generator / Create Random Numbers for Various Needs* 2026). Employee numbers were made through a random number generator (*RANDOM.ORG - String Generator* 2026). E-mail addresses for members were created based on their names and their membership type, for example, a student or professor would use their college e-mail address, a child would use their parents e-mail address. For staff members, the username and e-mail address had a specific format for the system, and passwords were created using an online "memorable" password generator (*Password Generator* 2026) and a SHA-256 calculator (emn178 2026).

2.3 Triggers

Two triggers were created for this project, one guaranteeing that a former employee's login is removed when an employee leaves, the other making so a book's availability status is updated when it is rented.

The first trigger, named "tg_deleteCredentialsWhenRemoveStaff" in the system, is fairly simplistic. When a record is deleted from the Staff table, that is, an employee record is removed, it triggers a DELETE operation that removes the Credentials entry associated to that record, that is, the username and password the (now former) employee used to use to log in.

The second trigger, named "tg_updateBookStatusOnRental" in the system, is a little more complex. Whenever a book is rented, that is, a new rental record is created associated with the book, or the status a book's existing/active rental record is updated, the trigger is responsible for executing an UPDATE action to reflect the state of that rental record on the book's availability status. When a rent record is created in order to put the book on hold, the book's status is reflected to "On Hold". When a rented record is created from a direct renting, or the status of a rented record is changed from "On Hold" to "On Time" (that is, the book was successfully lent), the book's status is changed to "Rented". When a book is returned, the book's status is changed to "Available", and when a book is lost while rented, the book's status is changed to "Unavailable for Renting".

2.4 Views

Two views were created for the system, one offering a wide-view of all the relevant information pertaining to a book, and one offering the number of books from each publisher contained in the library.

The first view, named "vw_LibraryCatalogue" in the system, offers a comprehensive look of each book and its pertinent information in a single view. It offers the book title, barcode, genre, publisher, language, availability status, publication date and the book's "blurb" description. This view offers the information that would be immediately of import when looking up a book in the library's system.

The second view, named "vw_PublisherBookStatistics" in the system, provides a list of publishers, and the total number of books from each publisher present in the library.

2.5 Procedures

Eight views were created for the system:

The first, named "sp_RentBook" in the system, handles the process of renting a book, that is, creating a new "RentRecord" entry. It takes the book being rented, by its barcode, the member renting the book, by ID, the library employee handling the rental, by employee ID, the status, whether it's being rented or put on hold, and the number of days the book will stay reserved in its current state, that is, the deadline for the book's devolution, in case it has been rented, or for the book to be checked out, if it has been put on hold. When creating the new rental record, it sets the rental date as the current date, and the due date as a number of days in the future equal to the number of days reserved, passed as a parameter.

The second, named "sp_ChangeRentalDueDate" in the system, adds a number of days to the rental record's due date, based on the currently registered due date. It takes a rental record by ID and a number of days, and adds those days to the current due date, updating the record with the new due date. Furthermore, if the beook is currently overdue, but the new due date is a day after the current date, it updates the record status to 'On Time'.

The third, named "sp_UpdateRentalStatus" in the system, is fairly simple, it simply updates the status of an existing rent record. Like the second procedure, it takes the rental record's ID and the new status to be updated to, and it only performs the UPDATE procedure on the rent record, updating it to the new status.

The fourth, named "sp_DeleteMember" in the system, deletes a member's records from the system and warns on whether the member has outstanding rentals or not. It takes the member's ID, verifies if there are any rental records under that member that have not yet been returned, printing a warning message if there are, and deletes the member's record from the system.

The fifth, named "sp_UpdateBookDescription" in the system, changes the description "blurb" of a book. It takes the book, by its barcode, and the new blurb text, then changes the book's XML "Description" field, substituting the old blurb for the new one, and updates the record.

The sixth, named "sp_PublisherBookCount" in the system, counts the number of books by publisher, and checks from which publishers the library has a books above a certain amount. It takes only that minimum number against which the number of books by each publisher is compared against.

The seventh, named "sp_SearchBooksByKeyword" in the system, searches in the system for any books containing a certain keyword, either in its title or description/"blurb". It takes only the keyword to be searched, and compares it against the title and description of all books in the system, returning a generalized/summary information similar to the summary provided by the "vw_LibraryCatalogue" view.

The eight, named "sp_BookAuthorXML" in the system, collates a book's information and creates from it a XML document. It takes a book's barcode and collects its title and the full name of its author, using the internally defined function "fn_FullAuthorName" (which itself returns the full name, divided by a space, of an author, given the author's ID in the system), and compiles the collected information into a XML document.

3 Conclusion

In conclusion, strong database design lends itself to easier and smoother implementation, however, without, and sometimes even with, in-depth knowledge of the system the database is being designed for, the domain it pertains to, and information tangential to its workings, unexpected aspects can surface during its implementation that necessitate for adjustment in design.

Moreover, the implementation of the system itself is not the end of its development cycle, and, whether an early prototype or a working system, there is often room for improvement, some of which can only be known once users are frequently using the system, and their needs and data pathways become understood.

For the system designed in this project, however, there are certain very clear improvements that could be made to improve the system:

3.1 Innovation

- Procedures to add new books, change aspects of books beyond the description, add new authors, add new publishers, new members, etc; all things that can be done with simple "raw" SQL queries, however, using procedures instead of direct queries help protect data consistency and helps divide user permission.
- On that note, actually implementing a system of login for all the staff users and permission control, allowing only users with certain permissions, as described on the "Credentials" table's user permissions, to perform certain actions.
- Adding the book's author's full name to the result of the view "vw_LibraryCatalogue" and stored procedure "sp_SearchBooksByKeyword".
- Implementing a system that only allows a certain order of rent record status changes, that is, for example, a book that is "On Time" may only become "Overdue", "Returned" or "Book Lost", it may never become "On Hold", or the record of a book that has been returned must not be changed to a different state.
- Performing some sort of automated warning when a book's due date is approaching, such as sending an e-mail to the member who rented it.
- Implementing a procedure that allows for searching books not only by their registered title, but also by possible alternate titles as registered in the description XML field.
- A procedure that allows searching books by their author.

References

- Code 39 Barcode Generator* (2026). URL: <https://orcascan.com/tools/code-39-generator> (visited on 07/19/2026).
- Do I Have to Pay Fines If I Return Items Late?* (Mar. 11, 2022). Libraries Ireland Support powered by Dublin City Libraries. URL: <https://support.librariesireland.ie/hc/en-ie/articles/4471806864401-Do-I-have-to-pay-fines-if-I-return-items-late> (visited on 07/19/2026).
- emn178 (2026). *SHA256*. Online Tools. URL: <https://emn178.github.io/online-tools/sha256.html> (visited on 07/19/2026).
- Goodreads* (2026). Goodreads. URL: <https://www.goodreads.com/> (visited on 07/19/2026).
- LibraryThing | Catalog Your Books Online | LibraryThing* (2026). LibraryThing.com. URL: <https://www.librarything.com> (visited on 07/19/2026).
- List of Languages by Total Number of Speakers* (July 12, 2026). In: *Wikipedia*. URL: https://en.wikipedia.org/w/index.php?title=List_of_languages_by_total_number_of_speakers&oldid=1363834848 (visited on 07/19/2026).
- Password Generator* (2026). *Password Generator: Strong, Secure & Random | 1Password*. URL: <https://1password.com/password-generator> (visited on 07/19/2026).
- Phone Number Generator | Create Random Numbers for Various Needs* (2026). URL: <https://dialaxy.com/lookups/phone-number-generator/> (visited on 07/19/2026).
- RANDOM.ORG - String Generator* (2026). URL: <https://www.random.org/strings/> (visited on 07/19/2026).
- The Meaning and History of First Names - Behind the Name* (2026). URL: <https://www.behindthename.com/> (visited on 07/19/2026).
- The Meaning and History of Surnames - Behind the Name* (2026). URL: <https://surnames.behindthename.com/> (visited on 07/19/2026).